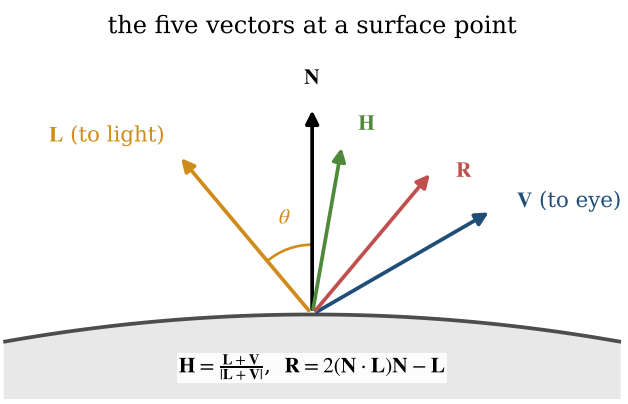
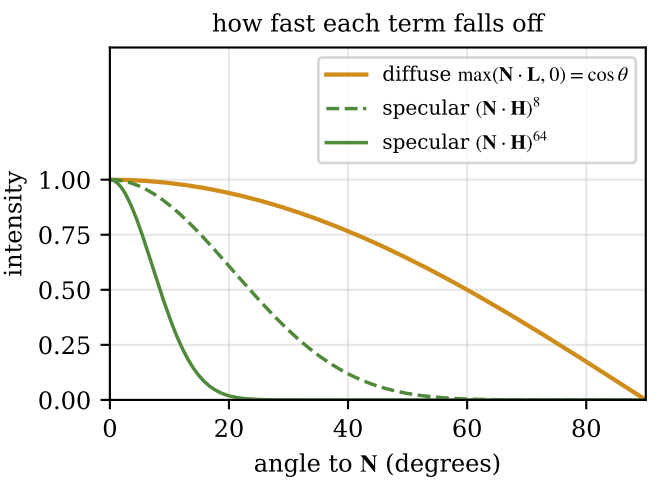

the five vectors at a surface point
N
L (to light)
H
R
V (to eye)
θ
$\mathbf{H} = \frac{\mathbf{L}+\mathbf{V}}{|\mathbf{L}+\mathbf{V}|}, \quad \mathbf{R} = 2(\mathbf{N}\cdot\mathbf{L})\mathbf{N} - \mathbf{L}$


how fast each term falls off
diffuse $\max(\mathbf{N}\cdot\mathbf{L}, 0) = \cos\theta$
specular $(\mathbf{N}\cdot\mathbf{H})^{8}$
specular $(\mathbf{N}\cdot\mathbf{H})^{64}$
intensity
1.00
0.75
0.50
0.25
0.00
0
20
40
60
80
angle to N (degrees)